# HOMEWORK NR. 1

Strength Analysis of a Bar Structure

Student code: 243483

## 1. Given Data

From the assignment tables:

$$F_1 = 32 \text{ kN}, \quad F_2 = 14 \text{ kN}, \quad F_3 = 52 \text{ kN}$$

$$\alpha = 55°, \quad \beta = 70°, \quad \gamma = 35°$$

Convert forces to Newtons:

$$F_1 = 32000 \text{ N}$$

$$F_2 = 14000 \text{ N}$$

$$F_3 = 52000 \text{ N}$$

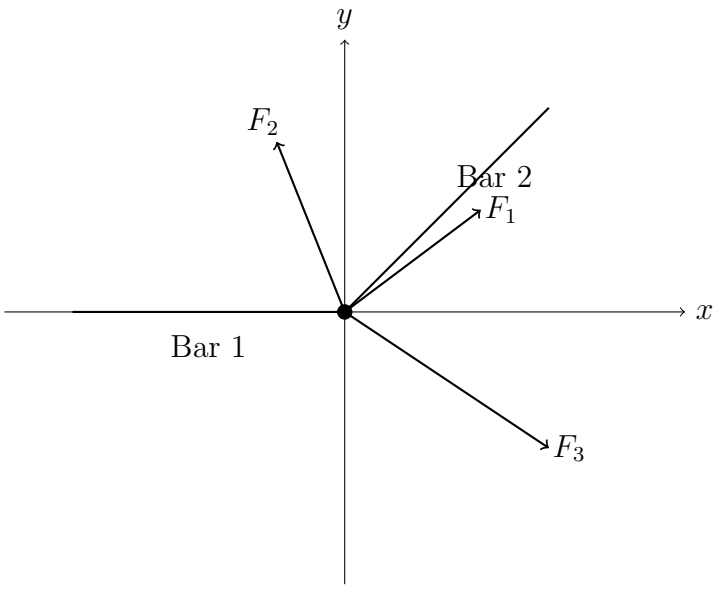
Material: Steel S355J2H

$$f_y = 355 \text{ MPa}$$

Allowable stress (safety factor 1.5):

$$\sigma_{allow} = 235 \text{ MPa}$$

## 2. Free Body Diagram



# 3. Resolve External Forces

### Force $F_1$

$$F_{1x} = F_1 \cos 55^\circ = 18355 \text{ N}$$
$$F_{1y} = F_1 \sin 55^\circ = 26214 \text{ N}$$

### Force $F_2$

$$F_{2x} = F_2 \cos 70^\circ = 4788 \text{ N}$$
$$F_{2y} = F_2 \sin 70^\circ = 13160 \text{ N}$$

### Force $F_3$

$$F_{3x} = F_3 \cos 35^\circ = 42588 \text{ N}$$
$$F_{3y} = F_3 \sin 35^\circ = 29848 \text{ N}$$

# 4. Equilibrium Equations

Let:

$$N_1 = \text{axial force in bar 1}$$
$$N_2 = \text{axial force in bar 2}$$

Components of $N_2$:

$$N_{2x} = N_2 \cos 35^\circ$$
$$N_{2y} = N_2 \sin 35^\circ$$

### Sum of Forces in $y$

$$\sum F_y = 0$$

$$0.574N_2 + 26214 + 13160 - 29848 = 0$$

$$N_2 = -16586 \text{ N}$$

$$\boxed{N_2 = 16.6 \text{ kN (compression)}}$$

### Sum of Forces in $x$

$$\sum F_x = 0$$

$$N_1 + 0.819(-16586) + 18355 + 4788 - 42588 = 0$$

$$\boxed{N_1 = 33.0 \text{ kN (tension)}}$$

# 5. Required Cross-Section Area

## Bar 1

$$A_1 = \frac{N_1}{\sigma_{allow}} = \frac{33031}{235 \times 10^6}$$

$$\boxed{A_1 = 141 \text{ mm}^2}$$

## Bar 2

$$A_2 = \frac{16586}{235 \times 10^6}$$

$$\boxed{A_2 = 70.6 \text{ mm}^2}$$

# 6. Selection of Standard Profile

Selected standard rectangular hollow section:

$$\boxed{RHS\ 40 \times 20 \times 2 \text{ (S355J2H)}}$$

Cross-sectional area:

$$A \approx 216 \text{ mm}^2$$

This satisfies the minimum required area for both bars.

# Final Results

$$N_1 = 33.0 \text{ kN (tension)}$$

$$N_2 = 16.6 \text{ kN (compression)}$$

$$A_{1,min} = 141 \text{ mm}^2$$

$$A_{2,min} = 70.6 \text{ mm}^2$$

Selected profile: RHS $40 \times 20 \times 2$